

Post-Hurricane Building Damage Detection from Satellite Imagery: A Comparison of CNN Architectures with Transfer Learning

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Abstract—Post-hurricane damage assessment from satellite imagery is an important task for emergency response, but manual inspection at city scale is too slow to be useful in the first days after a storm. We treat it as a binary image-classification problem (damaged vs. intact buildings) on the publicly available post-Hurricane-Harvey satellite tile dataset, and compare a small CNN trained from scratch with three ImageNet-pretrained backbones (ResNet-18, EfficientNet-B0, MobileNetV3-Small). Models are evaluated on a balanced 2000-tile test set and on a class-skewed 9000-tile test set with an 8:1 damage-to-intact ratio. EfficientNet-B0 reaches 99.55% accuracy and 0.9955 F1 on the balanced test, and 99.23% accuracy and 0.9957 F1 under the 8:1 skew. All three transfer-learning models pass 98.8% accuracy with at most 12M parameters. Grad-CAM overlays show the network’s attention sitting on damaged structures rather than on background pixels. Standard ImageNet-pretrained backbones with a replaced 2-class head therefore work well for this task and hold up under realistic class imbalance.

Index Terms—Hurricane damage assessment, satellite imagery, transfer learning, convolutional neural networks, Grad-CAM, class imbalance.

I. INTRODUCTION

Hurricane Harvey (2017) caused an estimated \$125 billion in damage in the United States, much of it concentrated in the Houston metropolitan area. In the days after a storm, emergency responders need to triage thousands of neighborhoods to allocate rescue, shelter, and clean-up resources. Doing this by ground inspection or by manually scanning aerial imagery is too slow, and the cost of a wrong call – both from missed damage and from wasted dispatches – is high. A model that classifies high-resolution satellite tiles can produce a damage map within hours.

We treat the task as a per-tile binary classification problem: each 128×128 RGB satellite tile is labeled either `damage` or `no_damage`. The dataset, collected after Hurricane Harvey and released on Kaggle, contains roughly 14k tiles split into a balanced training set, a balanced validation set, a balanced test set, and a second test set in which damaged tiles outnumber intact ones 8:1. The imbalanced split is included by the dataset authors to mimic field conditions, where most of the imagery in a flood-affected area shows some form of damage.

The contributions of this project are:

- Comparison of four architectures on the same training pipeline: a four-block CNN baseline trained from scratch

and three ImageNet-pretrained backbones (ResNet-18, EfficientNet-B0, MobileNetV3-Small).

- Evaluation on both the balanced and the 8:1 imbalanced test splits, reporting accuracy, precision, recall, F1 and ROC-AUC, so the behavior under class skew is measured rather than assumed.
- Grad-CAM overlays on the best model, including its small number of false-negative and false-positive cases, to check that the network is responding to plausible structural cues and not to background artifacts.

II. RELATED WORK

Convolutional neural networks have been the standard tool for satellite image classification since AlexNet [1]. Transfer learning from ImageNet is widely used in remote sensing because aerial datasets are usually small. ResNet [2] introduced residual connections that enable training of much deeper networks; EfficientNet [3] formulated a compound scaling rule for depth, width, and resolution; and MobileNetV3 [4] targeted edge deployment with depth-wise separable convolutions and squeeze-and-excitation blocks. The three backbones together cover a useful range of capacity (1.5M–11M parameters) and are common baselines in the remote-sensing literature.

For interpretability we use Grad-CAM [5], which produces class-discriminative heatmaps from the gradients of a target class with respect to a chosen convolutional feature map. It is the standard tool for checking what part of an image a CNN is responding to and is used here without modification.

For the specific Harvey tile dataset, Cao and Choe [8] showed that deep learning works well on this task, which is why we use the same publicly released splits.

III. DATASET

The dataset comes from Kaggle (kmader/satellite-images-of-hurricane-damage) and contains 128×128 RGB satellite tiles drawn from areas affected by Hurricane Harvey in 2017. Each tile has a binary label (`damage` or `no_damage`) based on visible structural damage to buildings.

The training and validation splits are balanced and used as-is, with no resampling. The two test sets are reported on separately: `test` is balanced and gives a clean comparison

TABLE I
DATASET SPLITS USED IN THIS WORK.

Split	damage	no_damage	Total
train_another	5,000	5,000	10,000
validation_another	1,000	1,000	2,000
test (balanced)	1,000	1,000	2,000
test_another (skew)	8,000	1,000	9,000

number, while `test_another` has an 8:1 ratio of damaged to intact tiles and serves as a stress test for class imbalance.

IV. METHODOLOGY

A. Models

Four convolutional architectures are compared, all set up for binary classification with a softmax over two logits.

SimpleCNN. A four-block CNN trained from scratch, used as a baseline to measure how much pretraining is worth on this task. Each of the first three blocks contains two 3×3 convolutions with batch normalization and ReLU, followed by 2×2 max pooling. The fourth block is a single convolution and adaptive average pooling, then a linear head with dropout. Total parameters: 436k.

ResNet-18 [2]. ImageNet-pretrained, with the original 1000-way fully connected layer replaced by a 2-way linear head. Parameters: 11.2M.

EfficientNet-B0 [3]. ImageNet-pretrained, with the final classifier replaced by a 2-way linear head. Parameters: 4.0M.

MobileNetV3-Small [4]. ImageNet-pretrained, with the final classifier replaced by a 2-way linear head. Parameters: 1.5M.

For all three pretrained models the entire network is fine-tuned (no backbone freezing). Head-only fine-tuning was tried first but converged to lower validation accuracy, so it is not reported.

B. Training Setup

All models are trained with the AdamW optimizer [6] (weight decay 10^{-4}) and a cosine learning-rate schedule [7] that decays from the initial value to zero over the training run. The from-scratch baseline uses a higher initial learning rate (10^{-3}) and is trained for eight epochs to give it more time to catch up. The three pretrained models use 3×10^{-4} and are trained for five epochs; their validation curves were already flat by epoch three (see Fig. 1).

Augmentation on the training split consists of random horizontal flips, random vertical flips, random rotations of up to $\pm 15^\circ$, and mild color jitter (brightness and contrast ± 0.15). Validation and test images receive only resizing and ImageNet-style normalization. Batch size is 128 throughout. The cross-entropy loss is computed per batch.

For each model we save the checkpoint with the highest validation accuracy and reload it for evaluation. All experiments run on a single NVIDIA RTX 3080. End-to-end training of all four models takes under twelve minutes.

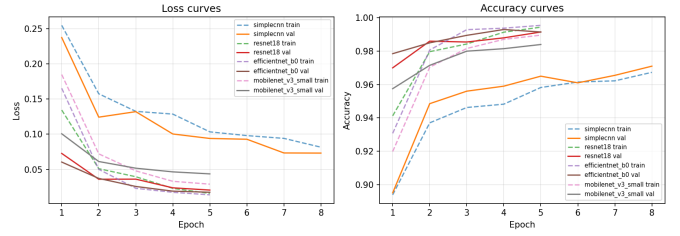


Fig. 1. Loss (left) and accuracy (right) on the training and validation splits for all four models. Pretrained backbones converge in 2–3 epochs; the from-scratch baseline needs all eight epochs to reach 97%.

C. Evaluation Protocol

Each model is evaluated on the balanced test split and on the 8:1 imbalanced split. We report accuracy, precision, recall, F1, and ROC-AUC, treating damage as the positive class. ROC-AUC is computed from the softmax probability of the positive class against a binarized target. Confusion matrices are produced for every model/split combination; for brevity the discussion below focuses on the best model.

D. Interpretability

For the best model (chosen by F1 on the balanced test split), we generate Grad-CAM [5] overlays on a small set of test tiles drawn from four groups: correct damage, correct no_damage, false negatives (true damage classified as intact), and false positives (true intact classified as damage). The target layer is the last convolutional block of the backbone; activation maps are normalized per-image and overlaid on the original RGB tile.

V. RESULTS

A. Training Dynamics

Fig. 1 shows the training and validation loss and accuracy curves for all four models. The three pretrained backbones reach a validation accuracy above 0.97 within a single epoch and plateau between epochs three and five. The from-scratch SimpleCNN starts at 0.89 validation accuracy and improves more gradually, finishing at 0.971 after eight epochs.

B. Balanced Test Set

Table II reports metrics on the 2000-tile balanced test set.

TABLE II
RESULTS ON THE BALANCED TEST SET ($n = 2000$). BEST PER COLUMN IN BOLD.

Model	Acc	Prec	Rec	F1	AUC
SimpleCNN	0.9705	0.9826	0.9580	0.9701	0.9963
ResNet-18	0.9920	0.9950	0.9890	0.9920	0.9997
EfficientNet-B0	0.9955	0.9940	0.9970	0.9955	0.9999
MobileNetV3-Small	0.9880	0.9929	0.9830	0.9879	0.9994

EfficientNet-B0 is best on every metric except precision, where ResNet-18 beats it by 0.001. The from-scratch baseline trails the best pretrained model by about 2.5 percentage points

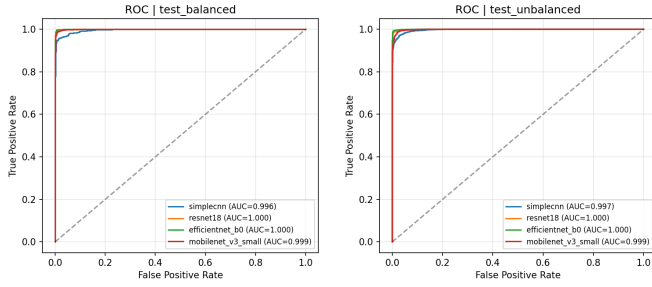


Fig. 2. ROC curves on the balanced (left) and 8:1 imbalanced (right) test sets. All four models have AUC above 0.996 on both splits.

of accuracy and F1, even though it was trained on the same data.

C. Class-Imbalanced Stress Test

Table III reports the same metrics on the 9000-tile imbalanced test set, which has an 8:1 majority of damaged tiles.

TABLE III
RESULTS ON THE IMBALANCED TEST SET ($n = 9000$,
DAMAGE:NO_DAMAGE = 8:1). BEST PER COLUMN IN BOLD.

Model	Acc	Prec	Rec	F1	AUC
SimpleCNN	0.9648	0.9966	0.9636	0.9799	0.9968
ResNet-18	0.9896	0.9990	0.9893	0.9941	0.9996
EfficientNet-B0	0.9923	0.9995	0.9919	0.9957	0.9997
MobileNetV3-Small	0.9834	0.9975	0.9839	0.9906	0.9987

All transfer-learning models lose only fractions of a percentage point moving from the balanced split to the imbalanced split. Precision goes up under skew (the minority class produces fewer false positives in absolute terms) while recall stays high. Fig. 2 shows the ROC curves on both test sets; AUC is at or above 0.997 for every pretrained model.

D. Confusion Matrix on the Imbalanced Split

Fig. 3 shows the EfficientNet-B0 confusion matrix on the 9000 imbalanced test tiles. Out of the 8000 truly damaged tiles, 7935 are classified correctly and 65 are missed. Of the 1000 intact tiles, 996 are classified correctly and 4 trigger false alarms. The false-alarm rate of $4/1000 = 0.004$ is the most relevant number if the system is used to dispatch responders.

E. Grad-CAM Analysis

Fig. 4 shows Grad-CAM overlays for EfficientNet-B0 on two correct damage predictions, two correct no_damage predictions, two false negatives, and two false positives drawn from the balanced test set.

On correct predictions the heatmap sits on the building footprint or on the obviously damaged area. The two false-negative cases are the more interesting ones: the network is still attending to the right structure, but the predicted-class probability is only 0.55–0.76. Lowering the softmax threshold below 0.5 (e.g. 0.35) would recover most of these missed cases at the cost of a few extra false alarms. The two false-positive

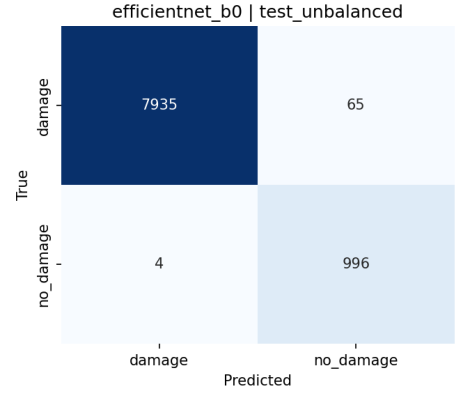


Fig. 3. Confusion matrix for EfficientNet-B0 on the 8:1 imbalanced test set. Out of 9000 tiles the model produces 65 false negatives and 4 false positives.

cases concentrate attention on small high-contrast features that look like debris but are in fact intact roofs.

F. Discussion

A few things are worth pointing out.

The 2.5-point gap between the from-scratch SimpleCNN and EfficientNet-B0 on a 10k-image dataset shows that ImageNet pretraining is still worth something for satellite tiles, even though the source domain (natural images) is quite different from the target. The pretrained features give a much stronger starting point than random initialization can reach in eight epochs.

The drop in accuracy between the balanced and imbalanced test sets is under 0.5 percentage points for every pretrained model, which is a sign that the classifiers are not relying on the training-time class prior. There is one caveat: the 8:1 majority class in the imbalanced set is the same as the trained-on positive class, which makes this an easier kind of imbalance than a flipped distribution would have been.

Parameter count is not a useful predictor of accuracy on this task. EfficientNet-B0 (4M) beats the larger ResNet-18 (11M), and is itself only 0.7 F1 points better than the much smaller MobileNetV3-Small (1.5M). The choice of backbone for deployment can therefore be made on memory and inference-speed grounds rather than on accuracy alone.

VI. CONCLUSION AND FUTURE WORK

We compared a from-scratch CNN baseline against three ImageNet-pretrained backbones for binary classification of post-Hurricane-Harvey satellite tiles. EfficientNet-B0 was the best model on every reported metric on both the balanced and the 8:1 imbalanced test split, with an F1 of 0.9955 on the balanced split and 0.9957 on the imbalanced one. Grad-CAM overlays showed the model’s attention on damaged structures rather than on background.

A natural follow-up would be to predict severity (e.g. minor, moderate, severe) instead of a binary label, which would need a relabeling pass. Building-level segmentation with a U-Net-style architecture would localize damage within a tile, which is

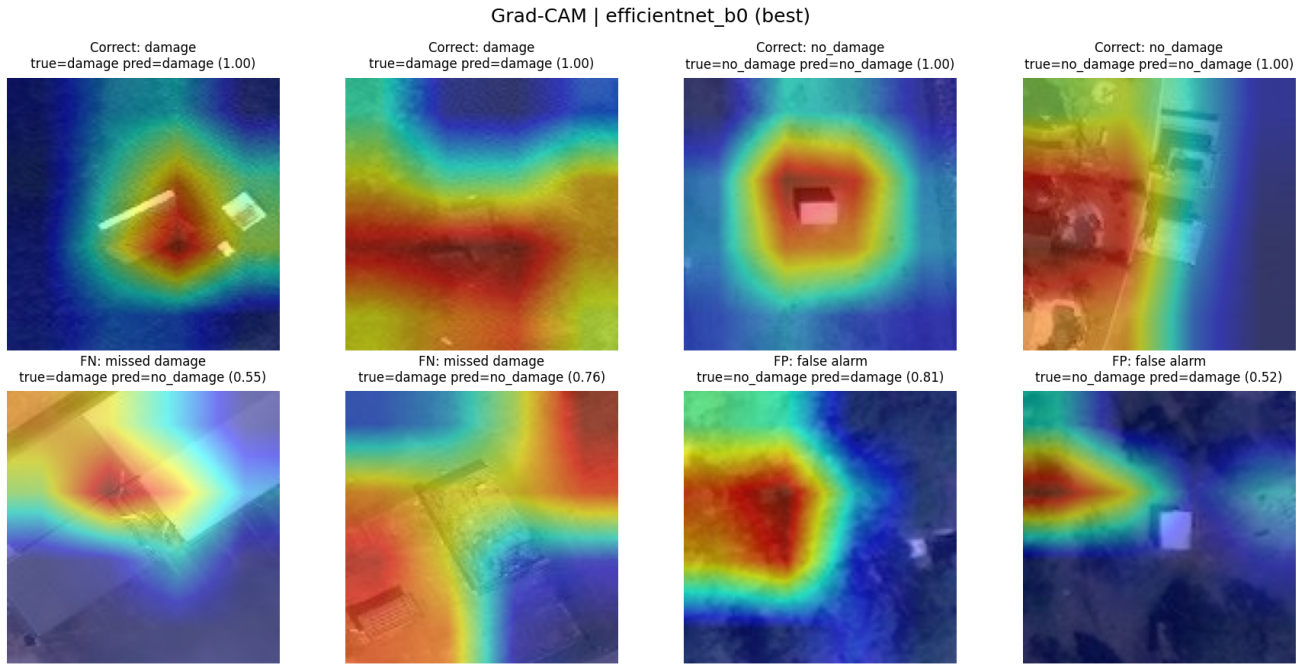


Fig. 4. Grad-CAM overlays on EfficientNet-B0. Top row: correct classifications. Bottom row: false negatives (the model still attends to the damaged structure but at lower confidence) and false positives (the model fixates on a small high-contrast feature). Numbers in parentheses are the predicted-class probabilities.

more useful operationally than a tile-level score. It would also be useful to test whether a Harvey-trained model generalizes to other hurricane datasets such as Maria or Irma without retraining. Finally, semi-supervised pretraining on unlabeled satellite imagery has been shown to close the gap between in-domain and ImageNet pretraining for some remote-sensing tasks; that would be a longer-term direction.

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